

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Nama Produk **OPTIZYME™ T4 DNA Polymerase**  
 Product Description: **OPTIZYME™ T4 DNA Polymerase**  
 Cat No. : BP8105-1, BP8105-5  
 Synonyms T4 DNA Pol

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

**Details of the supplier of the safety data sheet**

Company Fisher Scientific (M) Sdn Bhd No. 3, Jalan Sepadu 25/123,  
 Taman Perindustrian Axis, Seksyen 25,  
 40400 Shah Alam, Selangor Darul Ehsan, Malaysia.

Supplier .  
 E-mail address Enquiry.my@thermofisher.com

**Emergency Telephone Number**  
 (603) 5122 8888

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**
**Label Elements**

Signal Word Based on available data, the classification criteria are not met

Hazard Statements

Precautionary Statements

Other Hazards

**SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS**

| Component          | CAS-No    | Weight % |
|--------------------|-----------|----------|
| Water              | 7732-18-5 | 25 - 50  |
| Potassium chloride | 7447-40-7 | 1 - 2.5  |
| Glycerin           | 56-81-5   | >50      |

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## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

|                                           |                                                                                                                          |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.          |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.  |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Get medical attention.                                                                           |
| <b>Inhalation</b>                         | Remove to fresh air. Get medical attention immediately if symptoms occur. If not breathing, give artificial respiration. |
| <b>Self-Protection of the First Aider</b> | No special precautions required.                                                                                         |

### Most important symptoms and effects, both acute and delayed

No information available.

### Indication of any immediate medical attention and special treatment needed

**Notes to Physician** Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

No information available.

### **Hazardous Combustion Products**

None under normal use conditions.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Ensure adequate ventilation. Do not breathe dust/fume/gas/mist/vapors/spray.

### Environmental precautions

See Section 12 for additional Ecological Information. Should not be released into the environment.

### Methods and Material for Containment and Cleaning Up

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Soak up with inert absorbent material. Pick up and transfer to properly labelled containers.

## Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Ensure adequate ventilation. Wear personal protective equipment/face protection. Avoid breathing dust/fume/gas/mist/vapors/spray.

### Conditions for Safe Storage, Including any Incompatibilities

Keep container tightly closed. Store product at -20C.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component | Malaysia | ACGIH TLV | OSHA PEL                                                                                                                           |
|-----------|----------|-----------|------------------------------------------------------------------------------------------------------------------------------------|
| Glycerin  |          |           | (Vacated) TWA: 10 mg/m <sup>3</sup><br>(Vacated) TWA: 5 mg/m <sup>3</sup><br>TWA: 15 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup> |

  

| Component | European Union | The United Kingdom                         | Germany                                                                                                                                               |
|-----------|----------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Glycerin  |                | TWA: 10 mg/m <sup>3</sup> 8 hr (mist only) | TWA: 200 mg/m <sup>3</sup> (8 Stunden). AGW<br>- exposure factor 2<br>TWA: 200 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 400 mg/m <sup>3</sup> |

### Exposure Controls

#### Engineering Measures

None under normal use conditions.

### Personal protective equipment

#### **Eye Protection**

Wear safety glasses with side shields (or goggles)

#### **Hand Protection**

Protective gloves

#### **Skin and body protection**

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### **Respiratory Protection**

No protective equipment is needed under normal use conditions

#### **Recommended Filter type:**

Particle filter

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**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice

**Environmental exposure controls** No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Appearance                              | Colorless                |                                   |
| Physical State                          | Liquid                   |                                   |
| Odor                                    | Slight                   |                                   |
| Odor Threshold                          | No data available        |                                   |
| pH                                      | 7.5                      |                                   |
| Melting Point/Range                     | No data available        |                                   |
| Softening Point                         | No data available        |                                   |
| Boiling Point/Range                     | No information available |                                   |
| Flash Point                             | Not applicable           | Method - No information available |
| Evaporation Rate                        | No data available        |                                   |
| Flammability (solid,gas)                | Not applicable           | Liquid                            |
| Explosion Limits                        | No data available        |                                   |
| Vapor Pressure                          | No data available        |                                   |
| Vapor Density                           | No data available        | (Air = 1.0)                       |
| Specific Gravity / Density              | No data available        |                                   |
| Bulk Density                            | Not applicable           | Liquid                            |
| Water Solubility                        | Miscible                 |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Component                               | log Pow                  |                                   |
| Glycerin                                | -1.76                    |                                   |
| Autoignition Temperature                | Not applicable           |                                   |
| Decomposition Temperature               | No data available        |                                   |
| Viscosity                               | No data available        |                                   |
| Explosive Properties                    | No information available |                                   |
| Oxidizing Properties                    | No information available |                                   |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

No.

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## Chemical Stability

Stable under normal conditions.

## Possibility of Hazardous Reactions

**Hazardous Polymerization** No information available.  
**Hazardous Reactions** No information available.

## Conditions to Avoid

No information available.

## Incompatible Materials

No information available.

## Hazardous Decomposition Products

None under normal use conditions.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Acute Toxicity

#### Toxicology data for the components

| Component          | LD50 Oral                 | LD50 Dermal          | LC50 Inhalation              |
|--------------------|---------------------------|----------------------|------------------------------|
| Water              | -                         | -                    | -                            |
| Potassium chloride | LD50 = 2600 mg/kg ( Rat ) |                      |                              |
| Glycerin           | 12600 mg/kg ( Rat )       | > 10 g/kg ( Rabbit ) | > 2.75 mg/L/4h ( Rat )(mist) |

#### Chronic Toxicity

##### **Carcinogenicity**

There are no known carcinogenic chemicals in this product

##### **Sensitization**

No information available

##### **Mutagenic Effects**

None known

##### **Reproductive Effects**

No information available

##### **Developmental Effects**

No information available

##### **Target Organs**

No information available.

## SECTION 12: ECOLOGICAL INFORMATION

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## Ecotoxicity effects

| Component          | Freshwater Fish                                                                                    | Water Flea                               | Freshwater Algae    | Microtox |
|--------------------|----------------------------------------------------------------------------------------------------|------------------------------------------|---------------------|----------|
| Potassium chloride | Lepomis macrochirus:<br>LC50: 1060 mg/L /96h<br>Pimephales promelas:<br>LC50: 750 - 1020 mg/L /96h | EC50: 825 mg/L/48h                       | EC50: 2500 mg/L/72h |          |
| Glycerin           | LC50: 51 - 57 mL/L,<br>96h static<br>(Oncorhynchus mykiss)                                         | EC50: > 500 mg/L, 24h<br>(Daphnia magna) |                     |          |

## Persistence and degradability

### Persistence

Miscible with water, Persistence is unlikely, based on information available.

## Bioaccumulative potential

; Bioaccumulation is unlikely

| Component | log Pow | Bioconcentration factor (BCF) |
|-----------|---------|-------------------------------|
| Glycerin  | -1.76   | No data available             |

## Mobility in soil

The product is water soluble, and may spread in water systems. . Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

## Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

#### **Waste from Residues/Unused Products**

Dispose of in accordance with local regulations

#### **Contaminated Packaging**

Empty containers should be taken to an approved waste handling site for recycling or disposal

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN1845  
Hazard Class 9  
Packing Group III  
Proper Shipping Name CARBON DIOXIDE, SOLID

### Road and Rail Transport

Not regulated

### IATA

UN-No UN1845  
Hazard Class 9  
Packing Group III  
Proper Shipping Name CARBON DIOXIDE, SOLID

### **Special Precautions for User**

No special precautions required

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## SECTION 15: REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories X = listed

| Component          | EINECS    | ELINCS | NLP | TSCA | DSL | NDSL | PICCS | ENCS | IECSC | AICS | KECL         |
|--------------------|-----------|--------|-----|------|-----|------|-------|------|-------|------|--------------|
| Water              | 231-791-2 | -      |     | X    | X   | -    | X     | X    | X     | X    | KE-3540<br>0 |
| Potassium chloride | 231-211-8 | -      |     | X    | X   | -    | X     | X    | X     | X    | KE-2908<br>6 |
| Glycerin           | 200-289-5 | -      |     | X    | X   | -    | X     | X    | X     | X    | KE-2929<br>7 |

### National Regulations

**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** (volatile organic compound)

### **Key literature references and sources for data**

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Revision Date** 07-Feb-2020  
**Revision Summary** Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health**

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## Disclaimer

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**End of Safety Data Sheet**